

Module: Capstone Project

Related work and data description

Assignment: SS2 – Part A

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Date: 07th October

Part A — Table 1

Publication 1

Field	Details
Title	Data-Driven Product Returns Prediction: A Cloud-Based Ensemble Selection Approach
Harvard reference	Heilig, L., Hofer, J., Lessmann, S. and Voß, S. (2016) 'Data-Driven Product Returns Prediction: A Cloud-Based Ensemble Selection Approach', <i>Proceedings of the 24th European Conference on Information Systems (ECIS)</i> , Istanbul, Turkey.
Link	"DATA-DRIVEN PRODUCT RETURNS PREDICTION: A CLOUD-BASED ENSEMBLE SELECTI" by Leonard Heilig, Julien Hofer et al.
Modelling	The study used an ensemble selection approach. The models included CART, linear SVM, logistic regression, multilayer perceptron, random forest and AdaBoost. The researchers used directed hill-climbing and bagging to create the different ensembles. The models were then compared using accuracy and AUC.
Preprocessing techniques	The authors mention that a lot of preprocessing was done before the modelling. The dataset included numerical, date and categorical variables, such as item size, colour, customer state and manufacturer. The researchers also tested different training sample sizes to see how the size of the dataset affected the prediction performance.
Datasets used	The study used the DMC 2014 competition dataset. The dataset contains e-commerce apparel orders where products were either returned or kept. The training dataset contains 481,092 observations and 14 variables.
Key takeaways	1. Ensemble methods such as Random Forest may improve return prediction performance and should be considered during model comparison for the STADIOalot project. 2. Using more training data can improve the performance of the prediction model. 3. A return prediction model can be used before or during the purchase process to help retailers reduce avoidable returns.

Part A — Table 2

Publication 2

Field	Details
Title	Return management in e-commerce firms: A machine learning approach to predict product returns and examine variables influencing returns
Harvard reference	Mishra, A. and Dutta, P. (2024) 'Return management in e-commerce firms: A machine learning approach to predict product returns and examine variables influencing returns', <i>Journal of Cleaner Production</i> , 477, 143802.
Link	Return management in e-commerce firms: A machine learning approach to predict product returns and examine variables influencing returns - ScienceDirect
Modelling	The study compared three machine learning algorithms: logistic regression, random forest and gradient boosting. The models were compared using ROC curves and AUC to determine which model performed better at predicting product returns.
Preprocessing techniques	The study prepared e-commerce transaction data for machine learning by selecting variables related to orders, products and payment methods. The researchers looked at the variables that could influence product returns and prepared the data so that the different classification algorithms could be compared.
Datasets used	An e-commerce industry dataset containing information about product orders and customer return behaviour.
Key takeaways	1. Different machine learning models can be compared to find a better model for predicting returns. 2. Product category, order information and transaction variables should be included during feature engineering because they are associated with return behaviour. 3. ROC-AUC can be useful for comparing classification models when predicting whether an order will be returned.

Part A — Table 3

Publication 3

Field	Details
Title	Understanding and predicting online product return behavior: An interpretable machine learning approach
Harvard reference	Duong, Q.H., Zhou, L., Van Nguyen, T. and Meng, M. (2025) 'Understanding and predicting online product return behavior: An interpretable machine learning approach', <i>International Journal of Production Economics</i> , 280, 109499.
Link	Understanding and predicting online product return behavior: An interpretable machine learning approach - ScienceDirect
Modelling	The study developed an interpretable machine learning approach using product information taken from customer reviews. Random forest was used to predict product return behaviour and identify product and review characteristics that were linked to a higher chance of a return.
Preprocessing techniques	The study used customer review data and extracted product attributes from the reviews. The dataset included different product categories, such as food and groceries, books, fashion, electronics and home appliances.
Datasets used	Amazon product review data covering five product categories, including food and groceries, books, fashion, electronics and home appliances.
Key takeaways	1. Product characteristics and customer review-related information may provide valuable features for identifying high-risk return items. 2. Different product categories may have different factors that influence returns. 3. An interpretable model can help explain why an item has a higher return risk instead of only providing a prediction.

Part A — Table 4: Public Dataset

Public Dataset

Field	Details
Name	DMC 2014 Product Returns Dataset
Harvard reference	Data Mining Cup (2014) <i>DMC 2014 Product Returns Dataset</i> . Data Mining Cup 2014.
Link	https://www.data-mining-cup.com/reviews/dmc-2014/
Number of instances	The training dataset contains 481,092 instances and 14 variables. A separate test dataset contains 50,078 instances and 13 feature columns, as the return target is not included in the test data.
Number of feature columns	The training data contains 13 input feature columns, with the return outcome used as the target variable.
Number of classes	2 classes — returned and not returned.
Target range	Not applicable because this is a classification problem.
Origin	The dataset comes from the Data Mining Cup 2014 competition and contains online apparel retail transaction data. The training dataset contains 481,092 observations and 14 variables, including order, product, customer and delivery information, as well as the return outcome. The separate test dataset contains 50,078 observations and 13 predictor variables because the return outcome is not included.

The published description of the DMC dataset states that there are 481,092 training observations and 14 variables. These include order ID, item price, order date, delivery date, item ID, size, colour, manufacturer ID, customer information and the binary return outcome.

The DMC files include orders_train.csv as the training dataset, with 481,092 instances and 14 attributes. The orders_class.csv file is the test dataset, with 50,078 instances and 13 attributes because the return outcome is not included in the test data.